

GOOD AND BAD REDUCTION: EXAMPLES, WITH SOLUTIONS

SANTIAGO ARANGO-PIÑEROS

An elliptic curve over $\mathbb{Q}_p$ has good, multiplicative or additive reduction, and multiplicative reduction is split or nonsplit [Sil09, §VII.5]. This worksheet runs through concrete curves of each kind: Problem 1.5 reads the type off an equation, Problem 2.1 counts the rational points of the singular reduction, and Problem 3.1 changes the type by twisting the curve or extending the field. Problem 4.1 (★) is for home: the curves Frey attached to solutions of $A + B = C$ have no additive reduction at all.

Standing notation. p is a prime and $v = v_p$ its valuation on $\mathbb{Q}_p$, extended to $\overline{\mathbb{Q}_p}$. An elliptic curve E over $\mathbb{Q}_p$ is given by a Weierstrass equation with coefficients in $\mathbb{Z}_p$, with discriminant Δ and invariants c_4, c_6 [Sil09, §III.1]. When the equation is minimal, E_p is its reduction modulo p , a Weierstrass cubic over $\mathbb{F}_p$, and E_p^{ns} is E_p minus its singular point, if it has one. We write

$$a_p := p + 1 - \#E_p(\mathbb{F}_p),$$

counting the singular point too.

How to use this. Allow about 15 minutes for each of Problems 1.5, 2.1 and 3.1. The one Magma block, in Problem 1.5, runs in seconds in the free [online calculator](#); everything else is by hand.

1. READING OFF THE TYPE

For an equation $y^2 = x^3 + a_2x^2 + a_4x + a_6$ (that is, $a_1 = a_3 = 0$) we will use

$$(1.1) \quad \Delta = -16(4a_2^3a_6 - a_2^2a_4^2 - 18a_2a_4a_6 + 4a_4^3 + 27a_6^2), \quad c_4 = 16(a_2^2 - 3a_4),$$

which follow from the formulas of [Sil09, §III.1]. The facts below hold for every prime p , and are used throughout.

What we may use

Let K be $\mathbb{Q}_p$ or a finite extension of it, with normalized valuation v_K , ring of integers $\mathcal{O}_K$ and residue field k , and let E be an elliptic curve over K given by a Weierstrass equation with coefficients in $\mathcal{O}_K$.

Fact 1.1 (Minimal equations). Any other Weierstrass equation for E has discriminant $u^{-12}\Delta$ and invariant $u^{-4}c_4$ for some $u \in K^\times$. So if $v_K(\Delta) < 12$ or $v_K(c_4) < 4$, the equation is minimal [Sil09, Remark VII.1.1].

Fact 1.2 (Singular Weierstrass cubics). A Weierstrass cubic over a field with $\Delta = 0$ has exactly one singular point over an algebraic closure. It is a node, with two distinct tangent lines, if $c_4 \neq 0$, and a cusp, with one tangent line, if $c_4 = 0$ [Sil09, Proposition III.1.4(a)].

Fact 1.3 (Reduction types). Suppose that the equation is minimal. Then E has good reduction if $v_K(\Delta) = 0$, multiplicative reduction (the reduction has a node) if $v_K(\Delta) > 0$ and $v_K(c_4) = 0$, and additive reduction (a cusp) if $v_K(\Delta) > 0$ and $v_K(c_4) > 0$. Multiplicative reduction is split

if the slopes of the tangent lines at the node lie in k , and nonsplit otherwise [Sil09, §VII.5 and Proposition VII.5.1].

Fact 1.4 (Extending the field). Let L be a finite extension of K with ramification index e , so that $v_L = e v_K$ on K . If E has good, respectively multiplicative, reduction, then so does E_L . And E_L has good reduction for some finite extension L of K if and only if $j(E) \in \mathcal{O}_K$ [Sil09, Propositions VII.5.4(b) and VII.5.5].

Problem 1.5 (Reading off the type).

Let $p \geq 5$, and let $\varepsilon \in \mathbb{Z}$ be a non-square modulo p . For each equation, decide whether it is minimal at p , and find the reduction type of the curve over $\mathbb{Q}_p$: good, split multiplicative, nonsplit multiplicative, or additive.

- (1) $y^2 = x^3 + x^2 + p$.
- (2) $y^2 = x^3 + \varepsilon x^2 + p$.
- (3) $y^2 = x^3 + p$.
- (4) $y^2 = x^3 + p^6$.
- (5) $y^2 = x^3 + x^2 + p^{12}$.
- (6) Here is Magma's opinion at $p = 7$ and $\varepsilon = 3$. For each curve it prints the valuation of the minimal discriminant, the Kodaira symbol of the reduction (I_0 for good reduction, I_n with $n \geq 1$ for multiplicative, anything else for additive), and the polynomial $1 - a_7T$, or $1 - a_7T + 7T^2$ in the case of good reduction.

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R<T> := PolynomialRing(Integers());
p := 7; eps := 3;                                // 3 is not a square modulo 7
Info := function(a)
    E := EllipticCurve(a); L := LocalInformation(E, p);
    return <L[2], L[5], R!EulerFactor(E, p)>;
end function;
Info([0, 1, 0, 0, p]);                             // <1, I1, -T + 1>
Info([0, eps, 0, 0, p]);                           // <1, I1, T + 1>
Info([0, 0, 0, 0, p^6]);                           // <2, II, 1>
Info([0, 0, 0, 0, p^12]);                          // <0, I0, 7*T^2 + 4*T + 1>
Info([0, 1, 0, 0, p^12]);                          // <12, I12, -T + 1>

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Check that it agrees with your answers. Note the values of a_7 at the curves with bad reduction; Problem 2.1 explains them.

Solution.

Here $a_4 = 0$, so (1.1) reads $\Delta = -16a_6(4a_2^3 + 27a_6)$ and $c_4 = 16a_2^2$. Since $p \geq 5$, the factor 16 is a unit.

(1) $\Delta = -16p(27p + 4)$ has $v(\Delta) = 1 < 12$, so the equation is minimal (Fact 1.1), and $c_4 = 16$ is a unit: multiplicative reduction (Fact 1.3). The reduction $y^2 = x^3 + x^2 = x^2(x + 1)$ has its node at $(0, 0)$, where the lowest-degree terms $y^2 - x^2$ give the tangent lines $y = \pm x$. Their slopes lie in $\mathbb{F}_p$: split.

(2) $\Delta = -16p(4\varepsilon^3 + 27p)$ has $v(\Delta) = 1$, since $p \nmid \varepsilon$, and $c_4 = 16\varepsilon^2$ is a unit: multiplicative. The tangent lines to $y^2 = x^2(x + \varepsilon)$ at $(0, 0)$ are $y^2 = \varepsilon x^2$, with slopes $\pm\sqrt{\varepsilon} \notin \mathbb{F}_p$: nonsplit.

- (3) $\Delta = -432p^2$ has $v(\Delta) = 2$, so the equation is minimal, and $c_4 = 0$: additive. The reduction $y^2 = x^3$ has a cusp at $(0, 0)$.
- (4) Now $\Delta = -432p^{12}$ and $c_4 = 0$, so Fact 1.1 does not apply, and the reduction $y^2 = x^3$ of this equation has a cusp. But the equation is not minimal: substituting $x = p^2x'$, $y = p^3y'$ and dividing by p^6 gives $y'^2 = x'^3 + 1$, with $\Delta = -432$, a unit. So the curve has good reduction, and the cusp was an artifact of a bad model. This is why reduction types are defined with a minimal equation.
- (5) $\Delta = -16p^{12}(27p^{12} + 4)$ has $v(\Delta) = 12$, but $c_4 = 16$ is a unit, so the equation is minimal by Fact 1.1: a scaling as in (4) would divide c_4 by u^4 and leave $\mathbb{Z}_p$. The reduction is the same as in (1): split multiplicative. Large $v(\Delta)$ alone does not make an equation non-minimal.
- (6) The valuations 1, 1, 2, 0, 12 and the symbols agree: I_1 , I_1 and I_{12} are multiplicative, II is additive, and Magma finds the good reduction of (4), with $v(\Delta) = 0$ for its minimal equation. At the bad primes $a_7 = 1, -1, 0, 1$ for split, nonsplit, additive and split reduction. At the good one, $a_7 = -4$.

2. COUNTING RATIONAL POINTS ON THE SINGULAR REDUCTION

Problem 2.1 (Counting with lines through the singular point).

Let E be an elliptic curve over $\mathbb{Q}_p$ with bad reduction, and let S be the singular point of its reduction E_p .

- (1) Show that $S \in E_p(\mathbb{F}_p)$, and that $S \neq O$.^a
- (2) Show that a change of variables over $\mathbb{F}_p$ moves S to $(0, 0)$ and puts E_p in the form

$$y^2 + a_1xy = x^3 + a_2x^2,$$

and that the tangent lines at S are the lines $y = \alpha x$ with $g(\alpha) = 0$, where $g(t) := t^2 + a_1t - a_2$.

- (3) Let $t \in \mathbb{F}_p$. Show that the line $y = tx$ meets E_p at S and, if $g(t) \neq 0$, at exactly one more $\mathbb{F}_p$ -rational point $P_t = (g(t), tg(t))$, and nowhere else in the affine plane. Show that $t \mapsto P_t$ is a bijection from $\{t \in \mathbb{F}_p : g(t) \neq 0\}$ onto $E_p^{\text{ns}}(\mathbb{F}_p) - \{O\}$.
- (4) Deduce that $\#E_p^{\text{ns}}(\mathbb{F}_p) = p + 1 - r$, where r is the number of roots of g in $\mathbb{F}_p$, and hence that

$$a_p = \begin{cases} 1 & \text{for split multiplicative reduction,} \\ -1 & \text{for nonsplit multiplicative reduction,} \\ 0 & \text{for additive reduction.} \end{cases}$$

Compare with the output in Problem 1.5(6).

- (5) Check (4) by hand for $y^2 = x^3 + x^2$ and $y^2 = x^3 + 2x^2$ over $\mathbb{F}_5$.

^aHint: what does the Frobenius automorphism of $\overline{\mathbb{F}}_p$ do to a singular point?

Solution.

- (1) By Fact 1.3 and Fact 1.2, E_p has exactly one singular point S over $\overline{\mathbb{F}}_p$. The Frobenius automorphism σ of $\overline{\mathbb{F}}_p$ maps singular points of E_p to singular points, because E_p and its partial derivatives have coefficients in $\mathbb{F}_p$. So $\sigma(S) = S$, and $S \in E_p(\mathbb{F}_p)$. The rational point $O = [0 : 1 : 0]$ is nonsingular: in the chart $Y = 1$ the equation is $z + a_1xz + a_3z^2 = x^3 + a_2x^2z + a_4xz^2 + a_6z^3$, whose linear part z does not vanish.

- (2) Write $S = (x_0, y_0)$ and substitute $x+x_0$ for x and $y+y_0$ for y ; this keeps the Weierstrass form and has coefficients in $\mathbb{F}_p$. In the new equation $f(x, y) = y^2 + a_1xy + a_3y - x^3 - a_2x^2 - a_4x - a_6 = 0$, the rational point $(0, 0)$ lies on the curve, so $a_6 = 0$, and is singular, so $a_4 = -\partial f/\partial x(0, 0) = 0$ and $a_3 = \partial f/\partial y(0, 0) = 0$. The tangent lines at $(0, 0)$ are given by the terms of lowest degree, $y^2 + a_1xy - a_2x^2$, which factor as $(y - \alpha x)(y - \alpha' x)$ with α, α' the roots of g .
- (3) Substituting $y = tx$ gives $t^2x^2 + a_1tx^2 = x^3 + a_2x^2$, that is, $x^2(g(t) - x) = 0$. So the affine points of E_p on the line are S and, if $g(t) \neq 0$, the point P_t ; if $g(t) = 0$, the line is tangent at S and meets E_p only there. P_t is nonsingular, since $P_t \neq S$. Conversely, a nonsingular affine point (x, y) has $x \neq 0$, since $x = 0$ forces $y^2 = 0$, which is S . So it lies on the line of slope $t = y/x$ through S , with $g(t) \neq 0$, and equals P_t . The two maps are inverse to each other.
- (4) By (3), $\#E_p^{\text{ns}}(\mathbb{F}_p) = 1 + (p - r)$, counting O . By Fact 1.3 and (2), $r = 2$ for split multiplicative reduction, $r = 0$ for nonsplit, and $r = 1$ for additive reduction, where g is a square. Adding S , $\#E_p(\mathbb{F}_p) = p, p + 2, p + 1$, so $a_p = 1, -1, 0$. This matches Problem 1.5(6). The three values are the numbers of $\mathbb{F}_p$ -points of $\mathbb{G}_m$, of a twisted multiplicative group and of $\mathbb{G}_a$: E_p^{ns} is isomorphic to one of these as a group [Sil09, Proposition III.2.5], [Poo08, §6.3].
- (5) For $y^2 = x^3 + x^2$ over $\mathbb{F}_5$, $g(t) = t^2 - 1$ has the roots ± 1 , and $P_t = (t^2 - 1, t^3 - t)$ for $t = 0, 2, 3$ gives $(4, 0)$, $(3, 1)$ and $(3, 4)$. With O and S there are 5 rational points, so $a_5 = 1$. For $y^2 = x^3 + 2x^2$, $g(t) = t^2 - 2$ has no roots, since 2 is not a square modulo 5, and $t = 0, \dots, 4$ gives $(3, 0)$, $(4, 4)$, $(2, 4)$, $(2, 1)$ and $(4, 1)$: with O and S , 7 rational points, so $a_5 = -1$.

3. CHANGING THE CURVE, CHANGING THE FIELD

Problem 3.1 (Twists and extensions).

Let $p \geq 5$. For $d \in \mathbb{Q}_p^\times$ and $E: y^2 = x^3 + a_2x^2 + a_4x + a_6$, the **quadratic twist** of E by d is

$$E^d: y^2 = x^3 + da_2x^2 + d^2a_4x + d^3a_6.$$

- (1) Show that $(x, y) \mapsto (dx, d^{3/2}y)$ is an isomorphism $E \rightarrow E^d$ over $\mathbb{Q}_p(\sqrt{d})$, and that $\Delta(E^d) = d^6\Delta(E)$, $c_4(E^d) = d^2c_4(E)$ and $j(E^d) = j(E)$.

From now on, $E: y^2 = x^3 + x^2 + p$ is the curve of Problem 1.5(1), with split multiplicative reduction.

- (2) Let $\varepsilon \in \mathbb{Z}$ be a non-square modulo p . Show that E^ε has nonsplit multiplicative reduction, and that $a_p(E^\varepsilon) = -a_p(E)$. Since $\mathbb{Q}_p(\sqrt{\varepsilon})$ is an unramified extension of $\mathbb{Q}_p$, what does this say about which features of the reduction are preserved by unramified extensions?
- (3) Show that $E^p: y^2 = x^3 + px^2 + p^4$ has additive reduction. Over which quadratic extension of $\mathbb{Q}_p$ does it acquire multiplicative reduction? Show that it has good reduction over no finite extension of $\mathbb{Q}_p$.
- (4) Let $E_3: y^2 = x^3 + p$, and let L be a finite extension of $\mathbb{Q}_p$ with ramification index e . Show that E_3 has good reduction over L if and only if $6 \mid e$.^a
- (5) Show that $y^2 = x^3 + px$ has additive reduction, and good reduction over L as in (4) if and only if $4 \mid e$.

^aHint: for “if”, write $p = \varpi^e w$ with ϖ a uniformizer of L and w a unit.

Solution.

- (1) If $y^2 = x^3 + a_2x^2 + a_4x + a_6$, then $(d^{3/2}y)^2 = d^3y^2 = (dx)^3 + da_2(dx)^2 + d^2a_4(dx) + d^3a_6$,

so the image lies on E^d , and $(x, y) \mapsto (x/d, d^{-3/2}y)$ is the inverse. This is a change of variables with $u = d^{-1/2}$, which multiplies Δ by $u^{-12} = d^6$ and c_4 by $u^{-4} = d^2$ [Sil09, Table 3.1]. Hence $j = c_4^3/\Delta$ is unchanged. (Or use (1.1): every term of Δ is multiplied by d^6 , and every term of c_4 by d^2 .)

(2) $E^\varepsilon: y^2 = x^3 + \varepsilon x^2 + \varepsilon^3 p$ has $v(\Delta) = v(\Delta(E)) = 1$ and $v(c_4) = 0$, so it is minimal with multiplicative reduction, and its reduction $y^2 = x^2(x + \varepsilon)$ has tangent slopes $\pm\sqrt{\varepsilon} \notin \mathbb{F}_p$: nonsplit. By Problem 2.1, $a_p(E) = 1$ and $a_p(E^\varepsilon) = -1$. Over the unramified extension $\mathbb{Q}_p(\sqrt{\varepsilon})$, whose residue field is $\mathbb{F}_{p^2}$, the two curves become isomorphic, and both have split multiplicative reduction there. So an unramified extension preserves the reduction type (good, multiplicative, additive) but not the distinction between split and nonsplit.

(3) By (1), $v(\Delta(E^p)) = 6 + 1 = 7 < 12$, so the equation is minimal, and $v(c_4(E^p)) = v(16p^2) = 2 > 0$: additive reduction; its reduction is $y^2 = x^3$. Over $\mathbb{Q}_p(\sqrt{p})$, $E^p \cong E$ by (1), and E keeps its multiplicative reduction by Fact 1.4. Finally $v(j) = 3v(c_4) - v(\Delta) = -1$ for E , and hence for E^p , so $j(E^p) \notin \mathbb{Z}_p$ and E^p has good reduction over no finite extension (Fact 1.4).

(4) Over L , $v_L(\Delta) = e v(-432p^2) = 2e$. If E_3 has good reduction over L , a minimal equation over L has $v_L(\Delta') = 0$, and $\Delta' = u^{-12}\Delta$ by Fact 1.1, so $2e = 12v_L(u)$ and $6 \mid e$. Conversely, if $6 \mid e$, write $p = \varpi^e w$ and substitute $x = \varpi^{e/3}x'$, $y = \varpi^{e/2}y'$. Dividing by ϖ^e gives $y'^2 = x'^3 + w$, with $\Delta = -432w^2$ a unit: good reduction. For instance $L = \mathbb{Q}_p(p^{1/6})$ works.

(5) Here $\Delta = -64p^3$ and $c_4 = -48p$, so $v(\Delta) = 3$ and $v(c_4) = 1$: minimal, with additive reduction. Over L , $v_L(\Delta) = 3e$, and the same argument gives good reduction if and only if $12 \mid 3e$, that is, $4 \mid e$: for “if”, substitute $x = \varpi^{e/2}x'$, $y = \varpi^{3e/4}y'$ to get $y'^2 = x'^3 + wx'$, with $\Delta = -64w^3$ a unit. Both E_3 and this curve have $j \in \mathbb{Z}_p$ (0 and 1728), as Fact 1.4 requires.

4. ★ FREY CURVES

This section is [Sil09, Exercise 8.23], with the conductor of [Mil20, Aside II.3.5]; the LMFDB [LMF25] is the answer key.

Problem 4.1 (Frey curves are semistable).

An elliptic curve over $\mathbb{Q}$ is **semistable** if it has good or multiplicative reduction at every prime. Let A, B, C be nonzero integers with $A + B = C$ and $\gcd(A, B, C) = 1$. Let

$$E: y^2 = x(x + A)(x - B).$$

- (1) Show that $\Delta = 16A^2B^2C^2$ and $c_4 = 16(A^2 + AB + B^2)$.
- (2) Let p be an odd prime dividing ABC . Show that the equation is minimal at p and that E has multiplicative reduction at p . Show that the reduction is split if and only if $-B$, A or B is a square modulo p , according as p divides A , B or C .
- (3) Suppose that $A \equiv 0 \pmod{16}$ and $B \equiv 3 \pmod{4}$. Show that the substitution $x = 4x'$, $y = 8y' + 4x'$ gives an equation with coefficients in $\mathbb{Z}$,

$$y'^2 + x'y' = x'^3 + \frac{A - B - 1}{4}x'^2 - \frac{AB}{16}x',$$

with discriminant $(ABC)^2/2^8$ and $c_4 = A^2 + AB + B^2$. Deduce that E has good reduction at 2 if $v_2(A) = 4$, and multiplicative reduction at 2 if $v_2(A) \geq 5$. So E is semistable.

- (4) The **conductor** of E is $N = \prod_p p^{f_p}$, where $f_p = 0$ if E has good reduction at p , $f_p = 1$ if it has multiplicative reduction, and $f_p \geq 2$ if it has additive reduction. Under the hypotheses of part (3), compute N in terms of A, B, C . Find the curves for $(A, B, C) = (16, -1, 15)$

and $(32, -5, 27)$ in the LMFDB, and check their conductors and their split and nonsplit primes against your answers.

- (5) Suppose that $a^\ell + b^\ell = c^\ell$ with $\ell \geq 5$ prime and a, b, c nonzero coprime integers. Show that, after reordering and changing signs, $(A, B, C) = (a^\ell, b^\ell, c^\ell)$ satisfies the hypotheses of part (3), and that the resulting curve has conductor $\prod_{p|abc} p$ and minimal discriminant $(abc)^{2\ell}/2^8$.

Solution.

(1) The roots of $x(x+A)(x-B)$ are 0, $-A$ and B , with differences A , B and $A+B=C$. For $y^2 = (x-e_1)(x-e_2)(x-e_3)$,

$$\Delta = 16 \prod_{i < j} (e_i - e_j)^2, \quad c_4 = 16(e_1^2 + e_2^2 + e_3^2 - e_1e_2 - e_1e_3 - e_2e_3),$$

which follow from (1.1) with $a_2 = -(e_1 + e_2 + e_3)$, $a_4 = e_1e_2 + e_1e_3 + e_2e_3$ and $a_6 = -e_1e_2e_3$. Here they give $\Delta = 16A^2B^2C^2$ and $c_4 = 16(A^2 + B^2 + AB)$.

(2) Since A, B, C are pairwise coprime ($A+B=C$ and $\gcd(A, B, C) = 1$), p divides exactly one of them. If $p \mid A$, then $c_4 \equiv 16B^2$; if $p \mid B$, then $c_4 \equiv 16A^2$; if $p \mid C$, then $B \equiv -A$ and $c_4 \equiv 16A^2$. In each case c_4 is a unit, so the equation is minimal (Fact 1.1) and the reduction is multiplicative (Fact 1.3). The reduction and the tangent lines at its node are:

- $p \mid A$: $y^2 = x^2(x-B)$, node $(0,0)$, tangents $y^2 = -Bx^2$;
- $p \mid B$: $y^2 = x^2(x+A)$, node $(0,0)$, tangents $y^2 = Ax^2$;
- $p \mid C$: $y^2 = x(x+A)^2$, node $(-A,0)$, tangents $y^2 = -A(x+A)^2$, and $-A \equiv B$.

So the slopes lie in $\mathbb{F}_p$ exactly when $-B, A$, respectively B , is a square modulo p .

(3) Substituting and dividing by 64, $(8y' + 4x')^2 = 4x'(4x' + A)(4x' - B)$ becomes $y'^2 + x'y' = x'^3 + \frac{A-B-1}{4}x'^2 - \frac{AB}{16}x'$. The coefficients are integers because $A-B-1 \equiv 0-3-1 \equiv 0 \pmod{4}$ and $16 \mid A$. This is a change of variables with $u = 2$, so $\Delta' = 2^{-12}\Delta = (ABC)^2/2^8$ and $c'_4 = 2^{-4}c_4 = A^2 + AB + B^2$ (Fact 1.1). Since A is even and B odd, c'_4 is odd, so the new equation is minimal at 2. Since B and C are odd, $v_2(\Delta') = 2v_2(A) - 8$: good reduction if $v_2(A) = 4$, and multiplicative reduction if $v_2(A) \geq 5$ (Fact 1.3). With (2), and good reduction at the odd primes not dividing ABC , E is semistable. (At odd p , the substitution has $u = 2$, a unit, so it does not change the reduction type there.)

(4) Every f_p is 0 or 1, and $f_p = 1$ exactly at the odd primes dividing ABC , and at 2 when $32 \mid A$. So N is the product of the odd primes dividing ABC , times 2 if $32 \mid A$. For $(16, -1, 15)$, $N = 15$: the curve is 15a3 in Cremona's labels, with split multiplicative reduction at 5 ($B = -1$ is a square modulo 5) and nonsplit at 3 (-1 is not a square modulo 3). For $(32, -5, 27)$, $N = 30$: the curve is 30a2, split at 3 ($B = -5 \equiv 1$), nonsplit at 5 ($A = 32 \equiv 2$ is not a square modulo 5), and multiplicative at 2.

(5) Exactly one of a, b, c is even, since they are coprime and $a^\ell + b^\ell = c^\ell$. Since ℓ is odd, $x \mapsto x^\ell$ commutes with signs, so the relation can be rewritten as $A+B=C$ with A the ℓ -th power of the even one: $a^\ell + b^\ell = c^\ell$ itself if a is even, and $c^\ell + (-b)^\ell = a^\ell$ if c is even. Then $v_2(A) \geq \ell \geq 5$. If $B \equiv 1 \pmod{4}$, replace (A, B, C) by $(-A, -B, -C)$. By (4) and (3), the conductor is the product of all primes dividing ABC , which is $\prod_{p|abc} p$, and the minimal discriminant is $(ABC)^2/2^8 = (abc)^{2\ell}/2^8$, an ℓ -th power up to 2^8 . Frey observed that such a curve would be very strange; Ribet proved that, if it were modular, its ℓ -torsion would come from

a modular form of level 2, and there is none. Wiles and Taylor–Wiles proved that semistable elliptic curves over $\mathbb{Q}$ are modular, so the curve, and the solution, cannot exist [Mil20, §V.9].

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